# Propositional Logic Evaluator

by

Aye Khin Khin Hpone (Yolanda Lim)

Student ID: st125970

st125970@ait.asia

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Instructor: Prof. Phan Minh Dung

Teaching Assistant: Akraradet Sinsamersuk

Asian Institute of Technology

School of Engineering and Technology

Thailand

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# ABSTRACT

This report presents the design and implementation of a Propositional Logic Evaluator using classical compiler front-end techniques. The system accepts propositional logic expressions composed of truth values (`t`, `f`) and binary connectives ($\wedge$, $\vee$), and produces two outputs: the evaluated truth value and an equivalent expression in prefix (Polish) notation. The implementation follows a four-stage pipeline — lexical analysis, LALR(1) parsing, abstract syntax tree (AST) evaluation, and prefix translation. A context-free grammar with declarative precedence and associativity rules resolves ambiguity, and the SLY parser-generator library ensures a direct correspondence between the formal grammar and the Python implementation. An interactive GUI built with PySide6 (Qt6) provides convenient expression input and result display. All six required test cases pass correctly.

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# CHAPTER 1

# INTRODUCTION AND LANGUAGE DESIGN

## 1.1 Background and Problem Statement

Propositional logic is a foundational component of mathematical logic and computer science, dealing with propositions that are either **true** or **false** and the connectives used to combine them [Ben-Ari, 2012]. This report presents a **Propositional Logic Evaluator** that accepts expressions composed of truth values (`t`, `f`) and logical connectives ($\wedge$, $\vee$), and produces two outputs: the **evaluated truth value** and an equivalent expression in **prefix (Polish) notation**. Parenthesised grouping overrides default precedence, and $\wedge$ has higher priority than $\vee$. For example, `t ∨ f ∧ f` evaluates to `t` with prefix notation `(∨ t (∧ f f))`.

The implementation applies compiler front-end techniques [Aho et al., 2006]: lexical analysis, LALR(1) parsing, AST construction, and tree-based evaluation. The GUI is built with PySide6 (Qt6 for Python) and supports both Unicode ($\wedge$, $\vee$) and ASCII (`&`, `|`) operator input. Negation ($\neg$), implication ($\rightarrow$), and biconditional ($\leftrightarrow$) are outside the scope of this assignment.

## 1.2 Propositional Logic

The language of propositional logic expressions is defined *recursively* (as specified in the assignment):

1. Any of the truth values `t` (true) and `f` (false) is a propositional logic expression.
2. If $B$ and $B'$ are propositional logic expressions, then $B \wedge B'$ and $B \vee B'$ are propositional logic expressions.

Additionally, parenthesised grouping is supported: if $B$ is an expression then $(B)$ is an expression, allowing the user to override default precedence.

Table 1.1. Truth tables for conjunction ($\wedge$) and disjunction ($\vee$)

| $p$ | $q$ | $p \wedge q$ | $p \vee q$ |
|---|---|---|---|
| t | t | t | t |
| t | f | f | t |
| f | t | f | t |
| f | f | f | f |

**Conjunction** ($\wedge$): The result is `t` only when *both* operands are `t`.

**Disjunction** ($\vee$): The result is `t` when *at least one* operand is `t`.

### 1.3 Alphabet and Token Specification

The input language is defined over the alphabet shown in Table 1.2. Each symbol or symbol class is associated with a token type produced by the lexical analyser.

Table 1.2. Token specification for the propositional logic language

| Token Name | Regular Expression | Matches | Description |
|---|---|---|---|
| `TRUE` | `t` | `t` | Boolean true literal |
| `FALSE` | `f` | `f` | Boolean false literal |
| `AND` | `[∧&]` | ∧ or `&` | Conjunction operator |
| `OR` | `[∨\|]` | ∨ or `\|` | Disjunction operator |
| `LPAREN` | `\(` | `(` | Left parenthesis |
| `RPAREN` | `\)` | `)` | Right parenthesis |

Both Unicode symbols ($\wedge$, $\vee$) and ASCII alternatives (`&`, `|`) are accepted. Whitespace (spaces and tabs) is silently ignored.

### 1.4 Context-Free Grammar

The syntax of the propositional logic language is defined by the following context-free grammar (CFG):

$$G = (V_N,\ V_T,\ P,\ S) \tag{1.1}$$

where:

- $V_N = \{$ statement, expr $\}$ — non-terminal symbols
- $V_T = \{$ `TRUE`, `FALSE`, `AND`, `OR`, `LPAREN`, `RPAREN` $\}$ — terminal symbols

- $S$ = statement — start symbol
- $P$ — production rules, defined below

The production rules are:

Table 1.3. Production rules of the context-free grammar

| Rule | Production | Description |
|---|---|---|
| (1) | statement $\rightarrow$ expr | Top-level entry point |
| (2) | expr $\rightarrow$ expr `OR` expr | Disjunction |
| (3) | expr $\rightarrow$ expr `AND` expr | Conjunction |
| (4) | expr $\rightarrow$ `LPAREN` expr `RPAREN` | Parenthesised grouping |
| (5) | expr $\rightarrow$ `TRUE` | True literal |
| (6) | expr $\rightarrow$ `FALSE` | False literal |

**Ambiguity.** Rules (2) and (3) make the grammar ambiguous because the expression `t` $\vee$ `f` $\wedge$ `f` admits two distinct parse trees: one where $\vee$ is the root (grouping as $\texttt{t} \vee (\texttt{f} \wedge \texttt{f})$) and one where $\wedge$ is the root (grouping as $(\texttt{t} \vee \texttt{f}) \wedge \texttt{f}$). This ambiguity is resolved by declaring operator **precedence** and **associativity** in the LALR(1) parser (Section 1.5).

### 1.5 Operator Precedence and Associativity

To disambiguate the grammar, precedence levels and associativity are declared as shown in Table 1.4.

Table 1.4. Operator precedence and associativity

| Precedence Level | Operator | Associativity |
|---|---|---|
| 1 (lower) | $\vee$ (OR) | Left |
| 2 (higher) | $\wedge$ (AND) | Left |

**Consequence:** $\wedge$ binds tighter than $\vee$. In the expression `t` $\vee$ `f` $\wedge$ `f`:

$$\texttt{t} \vee \texttt{f} \wedge \texttt{f} \equiv \texttt{t} \vee (\texttt{f} \wedge \texttt{f}) \neq (\texttt{t} \vee \texttt{f}) \wedge \texttt{f} \tag{1.2}$$

**Left associativity** means that chains of the same operator group from left to right:

$$\texttt{t} \vee \texttt{f} \vee \texttt{f} \equiv (\texttt{t} \vee \texttt{f}) \vee \texttt{f} \tag{1.3}$$

These declarations are passed to the LALR(1) parser generator, which uses them to resolve shift/reduce conflicts in the generated parse table.

### 1.6 Notation Systems

The evaluator translates infix expressions into **prefix (Polish) notation** [Łukasiewicz, 1957], where the operator precedes its operands. Table 1.5 compares the three standard notation systems.

Table 1.5. Comparison of notation systems

| Notation | Name | Example | Operator Position |
|---|---|---|---|
| Infix | Standard | `t ∧ f` | Between operands |
| Prefix | Polish | `(∧ t f)` | Before operands |
| Postfix | Reverse Polish | `t f ∧` | After operands |

The key advantage of prefix notation is that it eliminates the need for precedence rules — the grouping structure is fully explicit through parenthesisation:

$$\text{Infix: } \mathtt{t} \vee (\mathtt{f} \wedge \mathtt{f}) \implies \text{Prefix: } (\vee\ \mathtt{t}\ (\wedge\ \mathtt{f}\ \mathtt{f})) \tag{1.4}$$

# CHAPTER 2

# SYSTEM ARCHITECTURE

## 2.1 Compiler Front-End Pipeline

The evaluator follows a classical compiler front-end pipeline [Aho et al., 2006] consisting of three sequential phases: lexical analysis, syntactic analysis, and semantic processing (evaluation and translation). Figure 2.1 illustrates the overall data flow.

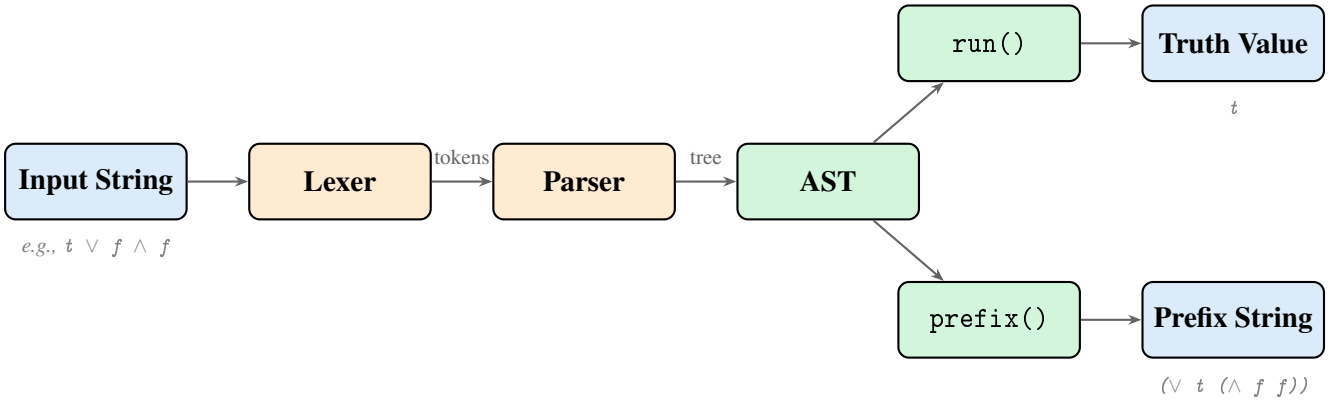



**Figure 2.1. System pipeline: input string → lexer → parser → AST → evaluation and translation**

Table 2.1 summarises the input, output, and responsible component for each phase.

Table 2.1. Phases of the compiler front-end pipeline

| Phase | Input | Output | Component |
|---|---|---|---|
| 1. Lexical Analysis | Character string | Token stream | `PropLogicLexer` |
| 2. Syntactic Analysis | Token stream | AST (tree) | `PropLogicParser` |
| 3a. Evaluation | AST | Boolean (`t`/`f`) | `Expression.run()` |
| 3b. Translation | AST | Prefix string | `Expression.prefix()` |

## 2.2 Project Structure

The source code is organised into a modular package structure under `src/125970eval/`:

Table 2.2. Project file structure and component responsibilities

| File | Responsibility |
|---|---|
| `main.py` | Entry point; GUI window and pipeline orchestration |
| `components/lexica.py` | Lexical analyser (tokeniser) |
| `components/parsers.py` | LALR(1) parser (AST builder) |
| `components/ui.py` | PySide6 GUI layout definition |
| `components/ast/statement.py` | AST node class hierarchy |

## 2.3 Technology Stack

Table 2.3 lists the tools and libraries used in this project.

Table 2.3. Technology stack

| Technology | Version | Purpose |
|---|---|---|
| Python | $\geq 3.10$ | Core implementation language |
| SLY [Beazley, 2023] | latest (GitHub) | Lexer and LALR(1) parser generator |
| PySide6 [The Qt Company, 2024] | $\geq 6.10.2$ | Qt6-based GUI framework |
| uv [Astral, 2024] | latest | Python package and virtual environment manager |

# CHAPTER 3

# IMPLEMENTATION DETAILS

## 3.1 Lexical Analyser

The lexical analyser is implemented in `components/lexica.py` as the class `PropLogicLexer`, extending `sly.Lexer`. It converts an input string into a stream of tokens using regular expression pattern matching.

```
class PropLogicLexer(Lexer):
    tokens = { TRUE, FALSE, AND, OR, LPAREN, RPAREN }

    ignore = ' \t'

    AND    = r'[∧&]'
    OR     = r'[∨|]'
    LPAREN = r'\('
    RPAREN = r'\)'
    TRUE   = r't'
    FALSE  = r'f'

    def error(self, t):
```

```
        self.index += 1
        print(f"ERROR: Illegal character '{t.value[0]}' "
              f"at line {self.lineno}")
```

Listing 3.1: Token definitions in `PropLogicLexer`

**Key design choices:**

- Both Unicode (∧, ∨) and ASCII (`&`, `|`) operators are accepted via character class regex patterns.
- The `ignore` attribute silently consumes whitespace.
- The `error()` method reports illegal characters with their line number, advances past the offending character, and continues scanning.

**Tokenisation example.** For the input `t ∨ f ∧ f`, the lexer produces the token stream shown in Table 3.1.

Table 3.1. Tokenisation of `t ∨ f ∧ f`

| Position | Character | Token Type | Token Value |
|---|---|---|---|
| 0 | t | TRUE | t |
| 2 | ∨ | OR | ∨ |
| 4 | f | FALSE | f |
| 6 | ∧ | AND | ∧ |
| 8 | f | FALSE | f |

Whitespace at positions 1, 3, 5, and 7 is consumed by the `ignore` directive.

## 3.2 Parser — LALR(1)

The parser is implemented in `components/parsers.py` as the class `PropLogicParser`, extending `sly.Parser`. It uses the **LALR(1)** (Look-Ahead Left-to-right Rightmost-derivation) parsing algorithm [Aho et al., 2006].

### *3.2.1 LALR(1) Parsing Algorithm*

LALR(1) is a **bottom-up** parsing algorithm that:

- Reads input **left to right** (L)
- Produces a **rightmost derivation** in reverse (R)

- Uses **1 token** of lookahead to make parsing decisions

It operates using a **shift-reduce** mechanism:

- **Shift:** Push the next input token onto the parse stack.
- **Reduce:** When the top of the stack matches the right-hand side of a production rule, replace it with the left-hand side non-terminal and execute the associated **semantic action** (constructing an AST node).

The SLY library automatically generates the LALR(1) parse table from the grammar rules and precedence declarations. A debug file (`parser.out`) is produced containing the full state machine for inspection.

### *3.2.2 Precedence Declaration*

```
precedence = (
    ('left', OR),       # Level 1 - lower priority
    ('left', AND),      # Level 2 - higher priority
)
```

Listing 3.2: Precedence declaration in `PropLogicParser`

In SLY, tokens listed **later** in the precedence tuple bind **tighter**. This ensures that $\wedge$ has higher precedence than $\vee$. When a shift/reduce conflict arises, the parser compares the precedence of the lookahead token with the precedence of the reduction rule and chooses accordingly.

### *3.2.3 Grammar Rules with Semantic Actions*

Each grammar rule is implemented as a decorated method that constructs the corresponding AST node:

```
@_('expr OR expr')
def expr(self, p) -> Expression:
    return Expression_logic(Operations.OR, p.expr0, p.expr1)

@_('expr AND expr')
def expr(self, p) -> Expression:
    return Expression_logic(Operations.AND, p.expr0, p.expr1)

@_('LPAREN expr RPAREN')
def expr(self, p) -> Expression:
    return p.expr

@_('TRUE')
```

```python
def expr(self, p) -> Expression:
    return Expression_bool(True)

@_('FALSE')
def expr(self, p) -> Expression:
    return Expression_bool(False)
```

Listing 3.3: Grammar rules with semantic actions

The semantic actions are **syntax-directed**: each grammar rule directly constructs the corresponding AST node, building the tree bottom-up as reductions occur during parsing.

### *3.2.4 Parsing Trace*

Table 3.2 shows the step-by-step LALR(1) shift-reduce parsing process for the input `t ∨ f ∧ f`, demonstrating how precedence resolves the ambiguity.

Table 3.2. LALR(1) shift-reduce parsing trace for `t ∨ f ∧ f`

| Step | Stack | Input | Action | Semantic Action |
|---|---|---|---|---|
| 1 | — | `t ∨ f ∧ f` | Shift `t` | — |
| 2 | `TRUE` | `∨ f ∧ f` | Reduce (5) | `Expression_bool(True)` |
| 3 | `expr` | `∨ f ∧ f` | Shift ∨ | — |
| 4 | `expr ∨` | `f ∧ f` | Shift `f` | — |
| 5 | `expr ∨ FALSE` | `∧ f` | Reduce (6) | `Expression_bool(False)` |
| 6 | `expr ∨ expr` | `∧ f` | Shift ∧ * | — |
| 7 | `expr ∨ expr ∧` | `f` | Shift `f` | — |
| 8 | `expr ∨ expr ∧ FALSE` | — | Reduce (6) | `Expression_bool(False)` |
| 9 | `expr ∨ expr ∧ expr` | — | Reduce (3) | `Expression_logic(AND, ...)` |
| 10 | `expr ∨ expr` | — | Reduce (2) | `Expression_logic(OR, ...)` |
| 11 | `statement` | — | Accept | Return root AST node |

* At step 6, a **shift/reduce conflict** occurs: the parser could either reduce `expr ∨ expr` by rule (2) or shift ∧. Because AND has **higher precedence** than OR, the parser chooses to **shift**, correctly binding ∧ tighter.

## 3.3 Abstract Syntax Tree (AST)

The AST is implemented in `components/ast/statement.py` using an inheritance hierarchy of three classes. Figure 3.1 shows the class structure.

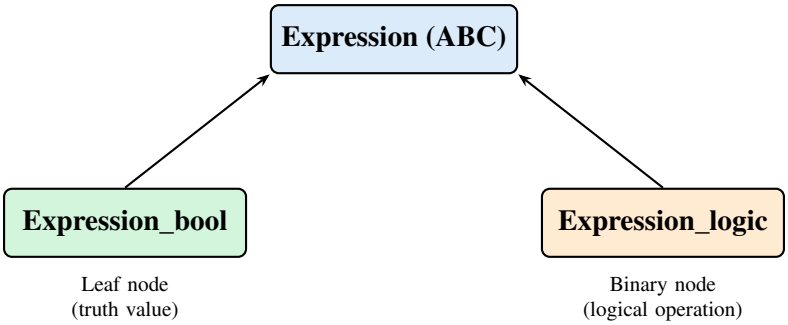



**Figure 3.1. AST class hierarchy**

Table 3.3 summarises each class.

Table 3.3. AST node classes

| Class | Role | Fields | Key Methods |
|---|---|---|---|
| `Expression` | Abstract base class | `value` | `run()`, `prefix()` (abstract) |
| `Expression_bool` | Leaf: truth literal | `value: bool` | `run()` → `bool`; `prefix()` → `"t"` or `"f"` |
| `Expression_logic` | Binary: logical op | `operation`, `left`, `right` | `run()` → recursive eval; `prefix()` → prefix string |

The `Operations` enum defines the two operations:

```
class Operations(Enum):
    AND = 0
    OR  = 1
```

Listing 3.4: `Operations` enum

### *3.3.1 AST Construction Example — Parse Tree*

For the input expression `t ∨ f ∧ f`, the parser constructs the AST shown in Figure 3.2. This tree is the direct result of the LALR(1) parsing process described in Table 3.2.

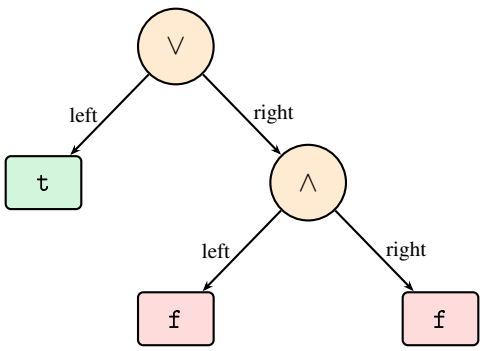



**Figure 3.2. AST (parse tree) for `t ∨ f ∧ f` — ∧ binds tighter and forms a subtree under ∨**

Figure 3.3 shows the AST for the parenthesised expression `(t ∨ f) ∧ f`, where parentheses override the default precedence.

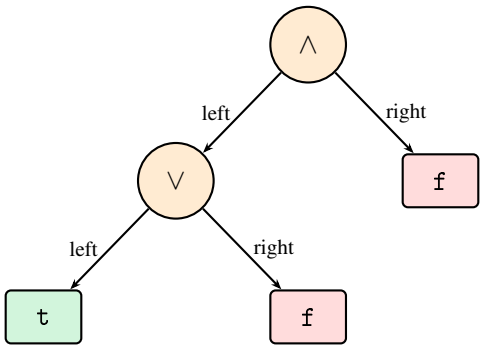



**Figure 3.3. AST (parse tree) for `(t ∨ f) ∧ f` — parentheses force ∨ to be evaluated first**

### 3.4 Evaluation — Truth Value Computation

The `run()` method evaluates the AST by performing a recursive **bottom-up (post-order) tree traversal**.

**For leaf nodes** (`Expression_bool`):

$$\texttt{run}() = \texttt{value} \in \{\text{True}, \text{False}\} \tag{3.1}$$

**For binary nodes** (`Expression_logic`):

$$\texttt{run}() = \begin{cases} \texttt{left.run()} \wedge \texttt{right.run()} & \text{if operation} = \text{AND} \\ \texttt{left.run()} \vee \texttt{right.run()} & \text{if operation} = \text{OR} \end{cases} \tag{3.2}$$

**Evaluation trace for** `t ∨ f ∧ f` (using the AST from Figure 3.2):

1. Evaluate left child of ∨: `Bool(t)` → True
2. Evaluate right child of ∨ (the ∧ subtree):
   (a) Evaluate left child of ∧: `Bool(f)` → False
   (b) Evaluate right child of ∧: `Bool(f)` → False
   (c) Compute: False ∧ False = False
3. Compute: True ∨ False = True → `t`

```python
def run(self):
    left_val = self.left.run()
    right_val = self.right.run()
    if self.operation == Operations.AND:
        self.value = left_val and right_val
    elif self.operation == Operations.OR:
        self.value = left_val or right_val
    return self.value
```

Listing 3.5: Evaluation method in `Expression_logic`

### 3.5 Translation — Prefix Notation Generation

The `prefix()` method converts the AST into a fully parenthesised prefix notation string via a recursive **pre-order tree traversal**.

**For leaf nodes:**

$$\texttt{prefix}() = \begin{cases} \texttt{"t"} & \text{if value is True} \\ \texttt{"f"} & \text{if value is False} \end{cases} \tag{3.3}$$

**For binary nodes:**

$$\texttt{prefix}() = \texttt{"("} + op + \texttt{" "} + \texttt{left.prefix()} + \texttt{" "} + \texttt{right.prefix()} + \texttt{")"} \tag{3.4}$$

where $op \in \{\wedge, \vee\}$ is determined by the node's operation.

**Translation trace for** `t ∨ f ∧ f`:

1. `prefix`($\vee$ node):
   (a) Left child: `prefix(Bool(t))` = `"t"`
   (b) Right child: `prefix`($\wedge$ node):
      i. Left: `prefix(Bool(f))` = `"f"`
      ii. Right: `prefix(Bool(f))` = `"f"`
      iii. Result: `"(∧ f f)"`
   (c) Result: `"(∨ t (∧ f f))"`

```python
def prefix(self) -> str:
    op_str = '∧' if self.operation == Operations.AND else '∨'
    return f"({op_str} {self.left.prefix()} {self.right.prefix()})"
```

Listing 3.6: Prefix translation in `Expression_logic`

**Key property:** The output is **fully parenthesised**, making the grouping structure explicit without relying on precedence rules. This ensures the prefix string is an unambiguous representation of the AST.

## 3.6 Graphical User Interface

The GUI is implemented using **PySide6** (Qt6 for Python) across two files: `components/ui.py` defines the layout and `main.py` implements the event-handling logic.

Figure 3.4 shows the GUI layout:

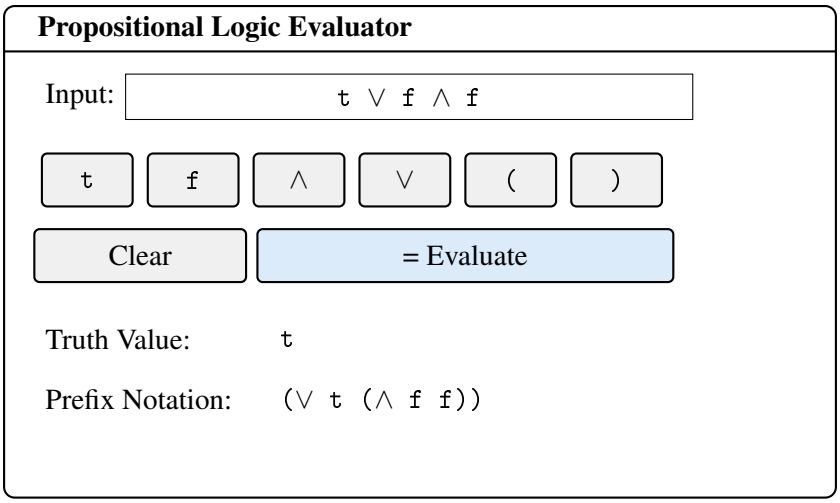
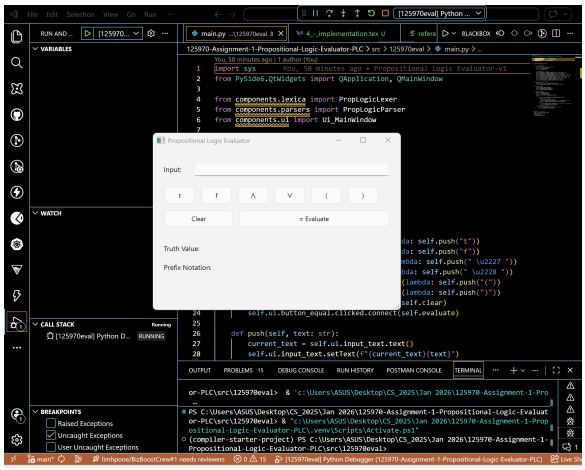
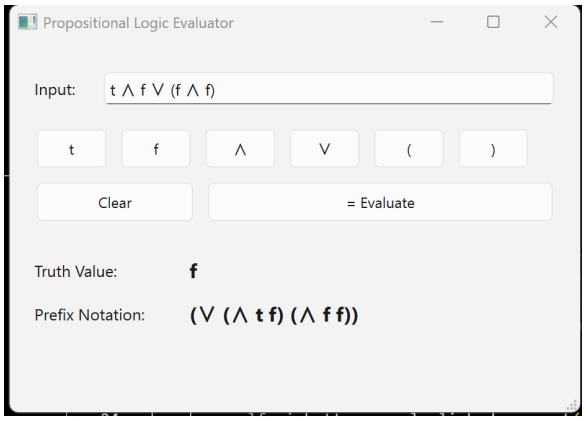



**Figure 3.4. GUI layout of the Propositional Logic Evaluator**



**(a) Input expression entered**




**(b) Evaluation result displayed**

**Figure 3.5. Screenshots — Application workflow: entering an expression (a) and viewing the evaluation result (b)**

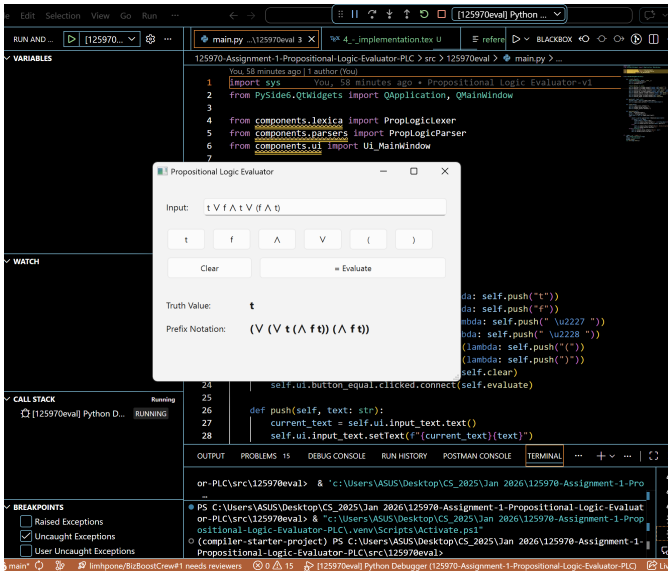



**Figure 3.6. Screenshot — Parenthesised expression demonstrating precedence override**

### *3.6.1 GUI Interaction Flow*

When the user clicks the **Evaluate** button, the following pipeline executes:

1. A fresh `PropLogicLexer` and `PropLogicParser` are instantiated.
2. The input text is tokenised by the lexer.
3. The parser builds an AST from the token stream.
4. `result.run()` computes the truth value.
5. `result.prefix()` generates the prefix notation string.
6. Both results are displayed in the output labels.
7. If the input is invalid, an error message is shown instead.

## 3.7 Integration of Parser and Translation

The assignment requires both a **truth value** and an **equivalent prefix notation** from a single input expression. A naïve approach would be to evaluate during parsing (embedding computation inside the grammar rules). Instead, this implementation uses the AST as an **intermediate representation** that decouples parsing from downstream processing. The integration works as follows:

1. **Parser → AST (syntax-directed translation).** Each grammar rule contains a **semantic action** (Listing 3.3) that constructs an AST node. As the LALR(1) parser performs bottom-up reductions, it simultaneously builds the tree — this is a form of *syntax-directed translation* [Aho et al., 2006]. The parser's output is the root `Expression` node.

2. **AST → Truth value (post-order traversal).** The root's `run()` method (Listing 3.5) recursively evaluates child nodes before the parent, implementing a bottom-up (*post-order*) traversal that mirrors the semantics of propositional logic.

3. **AST → Prefix string (pre-order traversal).** The root's `prefix()` method (Listing 3.6) visits the operator before its children, implementing a top-down (*pre-order*) traversal that directly produces fully parenthesised prefix notation.

**Why this design?** Because both `run()` and `prefix()` are independent traversals of the *same* AST, the expression is parsed **only once** and the tree is reused:

```
result = parser.parse(lexer.tokenize(input_text))  # parse once
truth_value = result.run()                          # traversal 1
prefix_str  = result.prefix()                       # traversal 2
```

Listing 3.7: Pipeline integration in `MainWindow.evaluate()`

Figure 3.7 summarises the data flow between the parser and the two translation passes.

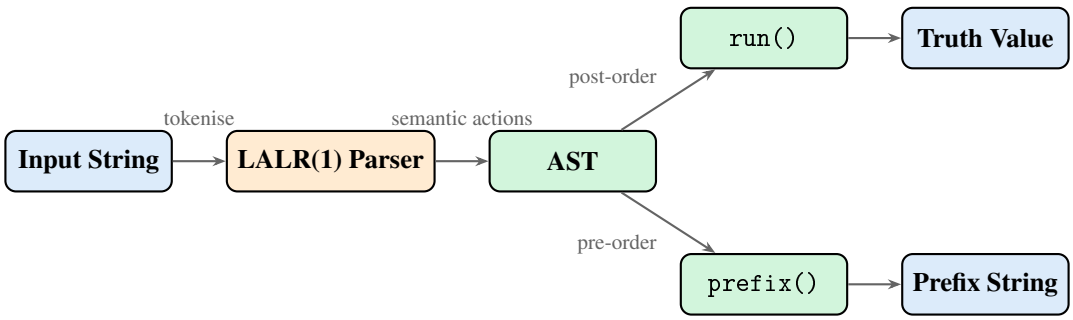



**Figure 3.7. Integration of parser and translation — the AST is the bridge between parsing and both output translations**

**Key insight:** The parser is responsible for *structural analysis* (building the tree with correct precedence and grouping), while translation is a separate phase that *walks* the already-built tree. This clean separation means adding a new output format (e.g. postfix notation) requires only a new traversal method — the parser and grammar remain unchanged.

# CHAPTER 4

# TESTING AND VERIFICATION

## 4.1 Test Cases

All six required test cases were executed and verified against expected outputs. Table 4.1 presents the complete results.

Table 4.1. Test case results — all cases pass

| # | Input | Expected Value | Actual Value | Expected Prefix | Actual Prefix |
|---|---|---|---|---|---|
| 1 | `t` | `t` | `t` | `t` | `t` |
| 2 | `f` | `f` | `f` | `f` | `f` |
| 3 | `t ∧ f` | `f` | `f` | `(∧ t f)` | `(∧ t f)` |
| 4 | `t ∨ f` | `t` | `t` | `(∨ t f)` | `(∨ t f)` |
| 5 | `t ∨ f ∧ f` | `t` | `t` | `(∨ t (∧ f f))` | `(∨ t (∧ f f))` |
| 6 | `(t ∨ f) ∧ f` | `f` | `f` | `(∧ (∨ t f) f)` | `(∧ (∨ t f) f)` |

**Result: All 6 test cases pass.** Expected and actual outputs match for both truth value and prefix notation.

## 4.2 Detailed Analysis of Key Test Cases

### *4.2.1 Test Case 5: `t ∨ f ∧ f` — Operator Precedence*

This test case validates that ∧ (AND) binds tighter than ∨ (OR).

**Without precedence rules** (left-to-right), the expression would be parsed as $(\mathtt{t} \vee \mathtt{f}) \wedge \mathtt{f} = \mathtt{f}$.

**With correct precedence** (∧ before ∨), the expression is parsed as $\mathtt{t} \vee (\mathtt{f} \wedge \mathtt{f}) = \mathtt{t}$.

The actual output `t` with prefix `(∨ t (∧ f f))` confirms that the parser correctly applies $\wedge > \vee$ precedence. The AST for this case is shown in Figure 3.2.

### *4.2.2 Test Case 6: `(t ∨ f) ∧ f` — Parenthesised Grouping*

This test case validates that explicit parentheses override the default precedence.

**Default precedence** would evaluate $\mathtt{f} \wedge \mathtt{f}$ first. But the parentheses force $\mathtt{t} \vee \mathtt{f}$ to be grouped first:

$$(\mathtt{t} \vee \mathtt{f}) \wedge \mathtt{f} \;=\; \mathtt{t} \wedge \mathtt{f} \;=\; \mathtt{f} \tag{4.1}$$

The actual output `f` with prefix `(∧ (∨ t f) f)` confirms correct parenthesised grouping. The AST for this case is shown in Figure 3.3.

### 4.3 Walkthrough: Complete Pipeline Execution

This section walks through the complete execution pipeline for the input `t ∨ f ∧ f`, demonstrating how each component transforms the data.

**Step 1 — Lexical Analysis:**

$$\texttt{t} \lor \texttt{f} \land \texttt{f} \longrightarrow \texttt{TRUE OR FALSE AND FALSE}$$

**Step 2 — Parsing (LALR(1) shift-reduce):** The parser processes the token stream as shown in Table 3.2 (Chapter 4), building the AST bottom-up through 11 shift/reduce steps.

**Step 3 — AST Construction:** The resulting AST is shown in Figure 3.2 (Chapter 4): a ∨ root with `t` as the left child and a ∧ subtree as the right child.

**Step 4 — Evaluation (`run()`):** Bottom-up traversal:

$$\texttt{f} \land \texttt{f} = \texttt{f} \tag{4.2}$$

$$\texttt{t} \lor \texttt{f} = \texttt{t} \tag{4.3}$$

**Step 5 — Translation (`prefix()`):** Pre-order traversal:

$$(\lor\ \texttt{t}\ (\land\ \texttt{f}\ \texttt{f})) \tag{4.4}$$

**Final Output:**

$$\text{Truth Value} = \texttt{t}, \quad \text{Prefix Notation} = \texttt{(}\lor\ \texttt{t (}\land\ \texttt{f f))}$$

### 4.4 Error Handling

The evaluator handles invalid input at three levels:

- **Lexical errors:** If the input contains an unrecognised character (e.g. `x`), the lexer's `error()` method reports the illegal character and its position, then advances past it to continue scanning.

- **Syntax errors:** If the token stream does not conform to the grammar (e.g. `t t` or `t` $\wedge$), the LALR(1) parser returns `None`, and the GUI displays an error message instead of a result.
- **Runtime exceptions:** Any unexpected exception during the pipeline is caught by the `evaluate()` method, and the error message is displayed in the truth value output field.

This layered approach ensures the application never crashes on malformed input and always provides feedback to the user.

# CHAPTER 5

# CONCLUSION

This report presented a **Propositional Logic Evaluator** that applies classical compiler front-end techniques to evaluate expressions over truth values (`t`, `f`) with conjunction ($\wedge$) and disjunction ($\vee$). The system follows a four-stage pipeline — lexical analysis, LALR(1) parsing (via SLY [Beazley, 2023]), post-order AST evaluation, and pre-order prefix translation — producing both the evaluated truth value and an equivalent prefix notation string from a single parse [Aho et al., 2006]. An interactive PySide6 GUI with dual Unicode/ASCII input modes completes the implementation. All six required test cases pass correctly (Chapter 4, Table 4.1).

Three design choices merit emphasis: using an **LALR(1) parser generator** keeps the grammar declarative and aligned with formal BNF notation; constructing an **AST as an intermediate representation** decouples parsing from downstream processing, enabling independent tree traversals for evaluation and translation without re-parsing; and producing **fully parenthesised prefix notation** ensures unambiguous output regardless of operator precedence. Future extensions could include additional connectives ($\neg$, $\rightarrow$, $\leftrightarrow$), named propositional variables, automatic truth table generation, and step-by-step evaluation visualisation.

# REFERENCES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Jan Łukasiewicz. *Aristotle's Syllogistic from the Standpoint of Modern Formal Logic*. Oxford University Press, Oxford, 2nd edition, 1957.

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# APPENDIX A: SOURCE CODE AVAILABILITY

The complete source code for the Propositional Logic Evaluator is publicly available on GitHub:

`https://github.com/limhpone/`
`125970-Assignment-1-Propositional-Logic-Evaluator-PLC.git`

The repository contains the following components:

- **Lexical analyser** (`components/lexica.py`) — tokenises input expressions into `TRUE`, `FALSE`, `AND`, `OR`, `LPAREN`, `RPAREN`
- **LALR(1) parser** (`components/parsers.py`) — parses token streams and builds an abstract syntax tree
- **AST node classes** (`components/ast/statement.py`) — `Expression_bool` (leaf) and `Expression_logic` (binary) with `run()` and `prefix()` methods
- **GUI layout** (`components/ui.py`) — PySide6 interface definition
- **Entry point** (`main.py`) — GUI window creation and pipeline orchestration

**How to Run**

1. Clone the repository:

```
git clone https://github.com/limhpone/
    125970-Assignment-1-Propositional-Logic-Evaluator-PLC.git
```

2. Install dependencies using `uv`:

```
uv sync
```

3. Run the application:

```
cd src/125970eval
uv run main.py
```

## Source Code Listings

The key implementation files are reproduced below for reference.

### *A.1 Lexical Analyser — components/lexica.py*

```python
from sly import Lexer

class PropLogicLexer(Lexer):
    tokens = { TRUE, FALSE, AND, OR, LPAREN, RPAREN }

    ignore = ' \t'

    AND    = r'[∧&]'
    OR     = r'[∨|]'
    LPAREN = r'\('
    RPAREN = r'\)'
    TRUE   = r't'
    FALSE  = r'f'

    @_(r'\n+')
    def ignore_newline(self, t):
        self.lineno += t.value.count('\n')

    def error(self, t):
        self.index += 1
        print(f"ERROR: Illegal character '{t.value[0]}' "
              f"at line {self.lineno}")
```

### *A.2 Parser — components/parsers.py*

```python
from components.lexica import PropLogicLexer
from components.ast.statement import (Expression, Expression_bool,
                                      Expression_logic, Operations)
from sly import Parser

class PropLogicParser(Parser):
    debugfile = 'parser.out'
    start = 'statement'
    tokens = PropLogicLexer.tokens
    precedence = (
        ('left', OR),
        ('left', AND),
    )

```

```python
    @_('expr')
    def statement(self, p):
        return p.expr

    @_('expr OR expr')
    def expr(self, p) -> Expression:
        return Expression_logic(Operations.OR, p.expr0, p.expr1)

    @_('expr AND expr')
    def expr(self, p) -> Expression:
        return Expression_logic(Operations.AND, p.expr0, p.expr1)

    @_('LPAREN expr RPAREN')
    def expr(self, p) -> Expression:
        return p.expr

    @_('TRUE')
    def expr(self, p) -> Expression:
        return Expression_bool(True)

    @_('FALSE')
    def expr(self, p) -> Expression:
        return Expression_bool(False)
```

### *A.3 AST Node Classes — `components/ast/statement.py`*

```python
from enum import Enum
from abc import ABC, abstractmethod

class Operations(Enum):
    AND = 0
    OR = 1

class Expression(ABC):
    @abstractmethod
    def __init__(self) -> None:
        self.value = None

    @abstractmethod
    def run(self):
        pass

    @abstractmethod
    def prefix(self) -> str:
        pass

class Expression_bool(Expression):
    def __init__(self, value: bool) -> None:
        self.value: bool = value

    def run(self):
        return self.value
```

```python

    def prefix(self) -> str:
        return 't' if self.value else 'f'

class Expression_logic(Expression):
    def __init__(self, operation: Operations,
                 left: Expression,
                 right: Expression) -> None:
        self.operation = operation
        self.left = left
        self.right = right
        self.value = None

    def run(self):
        left_val = self.left.run()
        right_val = self.right.run()
        if self.operation == Operations.AND:
            self.value = left_val and right_val
        elif self.operation == Operations.OR:
            self.value = left_val or right_val
        return self.value

    def prefix(self) -> str:
        op_str = '∧' if self.operation == Operations.AND else '∨'
        return (f"({op_str} {self.left.prefix()} "
                f"{self.right.prefix()})")
```

### *A.4 Main Entry Point — main.py*

```python
import sys
from PySide6.QtWidgets import QApplication, QMainWindow
from components.lexica import PropLogicLexer
from components.parsers import PropLogicParser
from components.ui import Ui_MainWindow

class MainWindow(QMainWindow):
    def __init__(self):
        super(MainWindow, self).__init__()
        self.ui = Ui_MainWindow()
        self.ui.setupUi(self)

        self.ui.button_t.clicked.connect(lambda: self.push("t"))
        self.ui.button_f.clicked.connect(lambda: self.push("f"))
        self.ui.button_and.clicked.connect(
            lambda: self.push(" \u2227 "))
        self.ui.button_or.clicked.connect(
            lambda: self.push(" \u2228 "))
        self.ui.button_lparen.clicked.connect(
            lambda: self.push("("))
        self.ui.button_rparen.clicked.connect(
            lambda: self.push(")"))
        self.ui.button_clear.clicked.connect(self.clear)
```

```python
        self.ui.button_equal.clicked.connect(self.evaluate)

    def push(self, text: str):
        current_text = self.ui.input_text.text()
        self.ui.input_text.setText(f"{current_text}{text}")

    def clear(self):
        self.ui.input_text.setText("")
        self.ui.output_value.setText("")
        self.ui.output_prefix.setText("")

    def evaluate(self):
        lexer = PropLogicLexer()
        parser = PropLogicParser()
        input_text = self.ui.input_text.text()
        try:
            result = parser.parse(lexer.tokenize(input_text))
            if result is not None:
                truth_value = result.run()
                self.ui.output_value.setText(
                    't' if truth_value else 'f')
                self.ui.output_prefix.setText(result.prefix())
            else:
                self.ui.output_value.setText(
                    "Error: Invalid expression")
                self.ui.output_prefix.setText("")
        except Exception as e:
            self.ui.output_value.setText(f"Error: {e}")
            self.ui.output_prefix.setText("")

if __name__ == "__main__":
    app = QApplication(sys.argv)
    window = MainWindow()
    window.show()
    sys.exit(app.exec())
```